

A POSITIVE ACCOUNT OF PROPERTY RIGHTS

BY DAVID FRIEDMAN

In thinking and talking about rights, including property rights, it seems natural to put the argument in either moral or legal terms. From the former viewpoint, rights are part of a description of what actions are right or wrong. The fact that I have a right to do something is an argument, although not necessarily a sufficient argument, that someone who prevents me from doing it is acting wrongly.¹

From the legal standpoint, rights are a description either of what the law says or of how it is enforced. On the latter interpretation, "I have a right to do X" translates as something like "If I do X the police will not arrest me; if someone tries to stop me from doing X the police will arrest him."² From this standpoint, one might claim that people in Holland have the right to buy marijuana and that people in America have the right to drive five miles per hour over the speed limit, even though both are illegal.

Both of these approaches have serious difficulties if our goal is to understand the phenomenon of rights, and associated phenomena, as they actually exist in the real world. We frequently observe behavior which looks like the claiming of rights and the recognition of rights in contexts where neither a moral nor a legal account seems relevant.

Consider, for example, Great Britain's "right" to control Hong Kong, Kowloon, and the New Territories. It is difficult to explain Communist China's willingness to respect that right on moral grounds, given that, from the Maoist standpoint, neither the government of Britain nor previous, non-Communist Chinese governments with which Britain had signed agreements were entities entitled to any moral respect. It seems equally difficult to explain it on legal grounds, given the general weakness of international law and the fact that for part of the period in question, Great Britain (as a member state of the United Nations) was at war with China. An alternative explanation—that the Chinese government believed that British occupation of Hong Kong was in its own interest—seems inconsistent with the Chinese failure to renew the lease on the New Territories, due to expire in 1997.

A second example is presented by the 1982 Falklands war. On the face of it, the clash looks like an attempted trespass repelled. Moral and legal

¹ For a sophisticated discussion of rights from the standpoint of moral philosophy, see Judith Jarvis Thomson, *The Realm of Rights* (Cambridge: Harvard University Press, 1990).

² Or the equivalent in civil rather than criminal terms.

accounts seem irrelevant, given the attitude of Argentina toward the British claim. Yet the willingness of Britain to accept costs far out of proportion to the value of the prize being fought over is difficult to explain except on the theory that the British felt they were defending their property, which raises the question of what that concept means in such a context.

A further difficulty with moral accounts of rights, in particular of property rights, is the degree to which the property rights that people actually respect seem to depend on facts that are morally irrelevant. This difficulty presents itself in libertarian accounts of property as the problem of initial acquisition. It is far from clear even in principle how unowned resources such as land can become private property. Even if one accepts an account, such as that of Locke, of how initial acquisition might justly have occurred, that account provides little justification for the existing pattern of property rights, given the high probability that any piece of property has been unjustly seized at least once since it was first cleared. Yet billions of people, now and in the past, base much of their behavior on respect for property claims that seem either morally arbitrary or clearly unjust.

A further difficulty with legal accounts of rights is that they are to some degree circular. We observe that police will act in certain ways and that their actions (and related actions by judges, juries, etc.) imply that certain people have certain rights. But the behavior of police is itself in part a consequence of rights—such as the right of the state to collect taxes and pay them to the police as wages, and the property right that the police then have over the money they receive.

For all of these reasons, I believe it is worth attempting a positive account of rights—an account which is both amoral and alegal. In Section I of this essay, I present such an account—one in which rights, in particular property rights, are a consequence of strategic behavior and may exist with no moral or legal support.³ The account is presented both

³The approach presented here is an extension of arguments made in my review of *Further Explorations in the Theory of Anarchy*, ed. Gordon Tullock (Blacksburg: University Publications, 1974), published in *Public Choice*, vol. 25 (1976), pp. 101–4. Elements of the argument have also appeared in David Friedman, "Many, Few, One—Social Harmony and the Shrunk Choice Set," *American Economic Review*, vol. 70, no. 1 (March 1980), pp. 225–32. A very similar approach is presented in Robert Sugden, *The Economics of Rights, Co-operation, and Welfare* (Oxford: Blackwell, 1986). The chief difference between Sugden's discussion and mine is that his argument is put primarily in terms of evolutionarily stable strategies, where mine is put in terms of Schelling points. Sugden also provides interesting examples of the same general line of argument in the work of eighteenth-century writers, in particular David Hume.

Throughout most of this essay, I present the argument as the working out of my own ideas. Readers familiar with the literature may feel, with some justice, that I fail to acknowledge how much of it has been said before by other people, often in somewhat different terms. I have presented it in this form in part because I am trying to work out the logic of a set of related ideas, not their history, and in part because that is how it in fact developed; most of the material in Section I of this essay was first written down, and some of it first published, more than fifteen years ago. I am grateful to the editors of this volume for providing an opportunity and incentive to compile and expand it.

as an explanation of how rights could arise in a Hobbesian anarchy and as an explanation of the nature of rights as we observe them around us. In Section II, I suggest ways in which something like the present structure of rights might have developed.

One puzzling feature of rights as we observe them is the degree to which the same conclusions seem to follow from very different assumptions. Thus, roughly similar structures of rights can be and are deduced by libertarian philosophers trying to show what set of natural rights is just and by economists trying to show what set of legal rules would be efficient. And the structures of rights that they deduce seem similar to those observed in human behavior and embodied in the common law. In Section III, I try to suggest at least partial explanations for this triple coincidence—the apparent similarity among what is, what is just, and what is efficient.

I. SCHELLING POINTS, SELF-ENFORCING CONTRACTS, AND THE PARADOX OF ORDER

Several writers have tried to analyze the transition from a Hobbesian state of nature to a state of civil order in terms of a set of hypothetical contracts establishing an initial distribution of property rights based on a pre-existing distribution of power.⁴ One difficulty with this approach is that in the initial situation there are no institutions to enforce contracts. How can people in that situation change it by making contracts which are unenforceable and thus of no effect?⁵

The same problem can be seen from the other side by asking in what sense our society, or any society, is ever *out* of a Hobbesian state of nature. What do we have, what have we created, that does not exist in the Hobbesian jungle? Civil order is not defined by the existence of physical objects—court rooms, police uniforms, law books. We can easily enough imagine a Hobbesian jungle—in the middle of a war, say—coexisting with all the physical appurtenances of civil society; and primitive peoples, without court rooms or law books, nonetheless live in a state of civil order.

Nor does it suffice to say that we are in a state of civil order because we have judges to interpret our laws and police to enforce them. Why do those people act in that way? Presumably because it is in their private interest to do so—just as potential criminals obey the law for the same reason. But that is how people act in the Hobbesian jungle. There too, one man may happen to enforce a rule, and another happen to obey it,

⁴ I would like to thank James Buchanan for bringing Schelling points to my attention and Gordon Tullock for provoking me into exploring them further. See especially Winston Bush, "Individual Welfare in Anarchy," in *Explorations in the Theory of Anarchy*, ed. Gordon Tullock (Blacksburg: University Publications, 1973).

⁵ This point is made by Gordon Tullock in "Corruption and Anarchy," in Tullock, ed., *Further Explorations*.

because each finds it in his own interest to do so. What is it that we have and the Hobbesian jungle does not have that makes it in the interest of people to behave in a law-abiding and peaceful manner? To say that the answer is "police, courts, government" only throws the question back a step; if civil order is enforced by men with guns, what controls them?

There are two sorts of answers to these questions. One is that the difference is a moral one. People somehow accept an obligation, agree *not* to behave according to simple self-interest, feel themselves bound by that agreement, and alter their actions accordingly.

There are difficulties with this sort of explanation. First, there is the empirical observation that people do not feel themselves bound to obey laws; many, perhaps most, people feel free to violate those laws (speed limits, drinking laws, customs regulations) which they disagree with and believe they can get away with breaking. Second, to the extent that people do feel a moral obligation to obey social rules, it is hard to derive that feeling from any variant of social-contract theory. The traditional variants encounter the difficulty eloquently described by Lysander Spooner:⁶ since we ourselves did not sign the contract, we are not bound by it.

The difficulties with deriving moral obligation from the sort of pairwise social contract suggested by Winston Bush⁷ are equally great. Even if we consider that each of us is, at every instant, in an implicit contract with each of his neighbors to respect some agreed-upon set of rights, still that contract, in Bush's model, is based on the threat of coercion. It has no more moral legitimacy, according to conventional moral ideas, than the obligation to pay off a protection racket.

It may be possible to explain the difference between a Hobbesian state of nature and civil society as a moral difference, but I prefer the alternative explanation—that the essential difference is not in the motivation of the players but in the strategic situation they face. This raises the question of how making an agreement—in a society with no mechanisms for enforcing agreements—can change anything, the strategic situation included.

A. *The tool: Schelling points*⁸

2,5,9,25,69,73,82,96,100,126,150

Two people are separately confronted with the list of numbers shown above and offered a reward if they independently choose the same num-

⁶ Lysander Spooner, *No Treason: No. VI, The Constitution of No Authority* (1870; reprint, Larkspur, CO: Pine Tree Press, 1966).

⁷ Bush, "Individual Welfare in Anarchy."

⁸ Thomas C. Schelling, *The Strategy of Conflict* (Oxford: Oxford University Press, 1960), ch. 3. While I am using Schelling's concept, my analysis of it, in particular my grounds for applying it to games with communication, is somewhat different from his.

ber. If the two are mathematicians, it is likely that they will both choose 2—the only even prime. Nonmathematicians are likely to choose 100—a number which seems, to the mathematicians, no more unique than the other two exact squares. Illiterates might agree on 69, because of its peculiar symmetry—as would, for a different reason, those whose interest in numbers is more prurient than mathematical.

There are three things worth noting about this simple problem in coordination without communication. The first is that each pair of players is looking for a number that is in some way unique. To a mathematician, all three squares are special numbers, as are the three primes. But if they try to coordinate on a square or a prime, they have only one chance in three of success—and besides, one may be trying primes and the other squares. The number 2 is unique. If the set of numbers did not contain 2 but did contain only one prime (or only one square, or one perfect number) they would choose that.

The second thing to note is that there is no single right answer; the number chosen by one player, and hence the number that ought to be chosen by the other, depends on the categories that the person choosing uses to classify the alternatives. The right strategy is to find some classification in terms of which there is a unique number, then choose that number—a strategy whose implementation depends on the particular classifications that pair of players uses. Thus, the right answer depends on subjective characteristics of the players.

The third point, which follows from this, is that it is possible to succeed in the game because of, not in spite of, the bounded rationality of the players. To a mind of sufficient scope every number is unique.⁹ It is only because the players are limited to a small number of the possible classification schemes for numbers, and because the two players may be limited to the same schemes, that a correct choice may exist. In this respect the theory of this game is radically different from conventional game theory, which assumes players with unlimited ability to examine alternatives and thus abstracts away from all subjective characteristics of the players except those embodied in their utility functions.¹⁰

Consider now two players playing the game called bilateral monopoly. They have a dollar to divide between them, provided they can agree on

⁹ There is a semi-serious theorem according to which all integers are interesting. The proof is by induction. If some positive integers are uninteresting, then there must be a smallest positive uninteresting integer. But this unique characteristic makes that number interesting. So there can be no smallest uninteresting positive integer, so there can be no uninteresting positive integers. Similarly, *mutatis mutandis*, for negative integers.

¹⁰ In practice, game theory sometimes smuggles subjective characteristics back into the argument in the process of choosing a particular strategy set. A famous example of this problem is the analysis of oligopoly. The assumption that the firm's strategy is defined as a choice of quantity and the assumption that it is defined as a choice of price lead to very different conclusions. See David Friedman, *Price Theory: An Intermediate Text* (Cincinnati: South-Western Publishing Co., 1990), ch. 11.

how to divide it. Superficially, there is no resemblance between this game and the one discussed above; the players are free to talk with each other as much as they want.

But while they can talk freely, there is a sense in which they cannot communicate at all. It is in my interest to persuade you that I will only be satisfied with a large fraction of the dollar; if I am really unwilling to accept anything less than ninety cents, you are better off agreeing to accept ten cents than holding out for more and getting nothing. Since it is in the interest of each of us to persuade the other of his resolve, all statements to that effect can be ignored; they would be made whether true or not. What each player has to do is to guess what the other's real demand is, what the fraction of the dollar is without which he will refuse to agree. That cannot be communicated, simply because it pays each player to lie about it. The situation is therefore similar to that in the previous game; the players must coordinate their demands (so that they add up to a dollar) without communication. It seems likely that they will do so by agreeing to split the dollar fifty-fifty.

The same points made about the previous game apply here, although less obviously. The players are looking for a unique solution; if I decide that the natural split is one-third/two-thirds, and you agree, both of us reasoning from a mystic belief in the significance of the number three, there is still the risk that each will decide he is entitled to the two-thirds.

To see that the solution depends on the particular categories used by the players, imagine that both have been brought up to believe that utility, not money, is the relevant payoff, and suppose further that both believe the marginal utility of a dollar to be inversely proportional to the recipient's income. In that case, the solution to the game is not a fifty-fifty split of money but a fifty-fifty split of utility—implying a division of the dollar into shares proportional to the two players' incomes.¹¹

Such an outcome, chosen because of its uniqueness, is called a Schelling point, after Thomas Schelling who originated the idea. It provides a possible solution to the problem of coordination without communication. As this example shows, it is relevant both to situations where communication is physically impossible and to situations where communication is impossible because there is no way that either party can provide the other with a reason to believe that what he says is true.

Even if it is impossible for the players in such a game to communicate their real demands, it may still be possible for them to affect the outcome by what they say. They could do so, not by directly communicating their own strategies (any such statement will be disbelieved), but by altering the other player's categories, the ways in which he organizes the alterna-

¹¹ I have discussed this point, and the game of bilateral monopoly, at greater length in David Friedman, "Bilateral Monopoly: A Solution," Fels Discussion Paper No. 52, University of Pennsylvania, March 1974 (unpublished, available from the author).

tives of the game, and thus changing the Schelling points which depend on those categories.

In the case just discussed, for example, one player (presumably the richer) might remind the other of their shared belief in the importance of utility in order to make sure the equi-utility Schelling point would be chosen. If, in the first game I described, the players were allowed to talk before seeing the numbers, a conversation on the interesting properties of primes or the special uniqueness of the lowest of a series of numbers might well alter the Schelling point, and thus the result of the game. One can interpret a good deal of bargaining behavior in this light—as an attempt by one party to make the other see the situation in a particular way, so as to generate a Schelling point favorable to the first party.

A slightly different way in which one may conceptualize the process of agreement on a Schelling point is in terms of bargaining costs in a context of continuous bargaining.¹² Consider a situation in which the number of possible outcomes is very large. Suppose the process of bargaining is itself costly, either because it consumes time or because each player bears costs (such as staying out on strike) in trying to validate his threats. As long as the players are faced with a choice among a large number of comparable alternatives, each proposal by one player is likely to call forth a competing proposal from another, slanted a little more toward his own interest.

But suppose there is one outcome that is seen as unique. A player who proposes that outcome may be perceived as offering, not a choice between that outcome, another slightly different, another different still, . . . but a choice between that outcome and continued bargaining. A player who says that he insists on the unique outcome and will not settle for anything less may be believable, where a similar statement about a different outcome would not be. He can convincingly argue that he will stand by his proposed outcome because, once he gives it up, he has no idea where he will end up or how high the costs of getting there will be.

In order for a Schelling point to provide a peaceful resolution to a conflict of interest, both parties must conceptualize the alternatives in similar ways—similar enough so that they can agree about which possible outcomes are unique, and thus attractive as potential Schelling points. So one interesting implication of the argument is that violent conflict is especially likely to occur on the boundary between cultures, where people with very different ways of viewing the world interact.

B. Up from Hobbes

Two people are living in a Hobbesian state of nature. Each can injure or steal from the other, at some cost, and each can spend resources on his

¹² This approach is discussed in Friedman, "Many, Few, One" (*supra* note 3).

own defense. Since conflict consumes resources, both could benefit by agreeing on what each owns and thereafter each respecting the other's property. The joint benefit might be divided in different ways, according to the particular set of property rights they agree on—what property belongs to whom, and whether either has a property right in tribute from the other. This is a special case of the game—bilateral monopoly—described above.

Each player, of course, will threaten to refuse to make any such agreement unless he gets the division he wants. Each will disbelieve most of the other's threats. If their ability to coerce and defend is roughly equal, and if there is some natural division of contested property (such as a stream running between their farms), it is likely that they will find a Schelling point in the form of an agreement to accept that division, respect each other's rights, and pay no tribute.

If one (being, perhaps, slightly more powerful) tries to insist on a small tribute, arguing that it will still leave the other better off than continued conflict, the other may believably refuse, arguing that once he concedes any tribute there is no natural limit to what the other can demand. Agreeing to tribute costs the victim not only the tribute but the only available Schelling point. The expected cost to him of such an agreement includes both the possible cost of paying higher tribute in the future and the risk of future conflicts if in the future he rejects demands for higher tribute. That cost may be high enough to establish the credibility of his insistence that he will choose continued conflict over the payment of even a small tribute.

So far we have considered the Schelling point that generates an agreement. But the agreement itself, whether generated by a Schelling point or in some other way, is thereafter itself a Schelling point. It is a unique outcome of which both players are conscious. Once it has been made, a policy of "if you do not abide by the agreement, I will revert to the use of force, even if the violation is small compared to the cost of conflict" is believable for precisely the same reason the refusal to pay tribute, or any insistence by a bargainer on a Schelling point, is believable. The signing of a contract establishes a new Schelling point and thereby alters the strategic situation. The contract enforces itself.

This applies not only to the initial pairwise social contract but to subsequent contracts as well. Suppose you have an orchard and I have an axe. After agreeing on our mutual property rights, you offer me a bushel of apples to cut down a tree that is shading your orchard. I cut down the tree as agreed, but you refuse to give me the apples. What happens?

So far as our physical situation is concerned, I am no more able to compel you to pay me a bushel of apples now than I was before you made the offer and I cut down the tree—our material resources, our ability to hurt each other and defend ourselves, are the same as they were. Yet my threat to cut down your orchard unless you pay up is more credible than it would have been before, both because I have more reason to carry

through on it and because you have less reason to resist it. Before, the attempt to get a bushel of apples from you would have been an attempt to move you away from the Schelling point established by the initial contract. Now it is an attempt to restore the Schelling point established by our subsequent agreement.

A more conventional explanation of this is that the reason it is in your interest to deliver the apples once you have agreed to do so is that you wish to establish a reputation for keeping promises, and that the reason it is in my interest to punish you if you do not deliver the apples is because I wish to establish a reputation for enforcing contracts made with me. While this may be true, there are two reasons why it cannot be a complete explanation. First, it depends on a particular perception of consistent behavior—in pure logic, there is no more reason to think of “always enforce” as more consistent than “back down the first, third, fifth, . . . time and fight the second, fourth, sixth, . . . time.” Each describes a single possible strategy. The important difference between them is that the former is a Schelling point and the latter is not—a fact not about the strategies but about the way we classify them.

A second and related problem with the conventional account is that I might equally well wish to establish a reputation as a consistent extortionist. We need some way of explaining why I cut down the shade tree first, instead of simply committing myself to demand your apples. If the former pattern creates a Schelling point of contract fulfillment and the latter does not, that provides a possible explanation.

I believe I have now resolved the apparent paradox of contracting out of the Hobbesian jungle. The process of contracting changes the situation because it establishes new Schelling points, which in turn affect the strategic situation and its outcome. The same analysis can be used from the other side to explain what constitutes civil society. The laws and customs of civil society are an elaborate network of Schelling points. If my neighbor annoys me by growing ugly flowers, I do nothing. If he dumps his garbage on my lawn, I retaliate—possibly in kind. If he threatens to dump garbage on my lawn, or play a trumpet fanfare at 3 A.M. every morning, unless I pay him a modest tribute, I refuse—even if I am convinced that the available legal defenses would cost more than the tribute he is demanding.

If a policeman arrests me—even for a crime I did not commit—I go along peacefully. If he tries to rob my house, I fight, even if the cost of doing so is more than the direct cost of letting him rob me. Each of us knows what behavior by everyone else is within the rules and what behavior implies unlimited demands, the violation of the Schelling point, and the ultimate return to the Hobbesian jungle. The latter behavior is prevented by the threat of conflict even if (as in the British defense of the Falklands) the direct costs of surrender are much lower than the direct costs of conflict.

One question this raises is how we succeed in committing ourselves not

to back down in such situations. One answer has been suggested already. It is in my long-run interest not to back down because if I do I can expect further demands: "[I]f once you have paid him the danegeld/You never get rid of the Dane."¹³

This explanation is not entirely adequate. In some situations, the aggressor may be able to commit himself to keeping your surrender secret and limiting his own demands. In others, the short-run costs of resistance may be larger than the long-run costs of surrender.

People (and nations) do sometimes surrender to such demands. If they do so less often than a simple calculation of costs and benefits might predict, the explanation may be found in a class of arguments made by Robert Frank and others.¹⁴

The central insight of such arguments is that even if surrender is sometimes in my private interest, being the sort of person who will surrender when it is in his interest to do so may not be, since if it is known that I will not back down, there is no point in making the initial demand.¹⁵ My first best option is to pretend to be tough, in the hope that the demand will not be made, while reserving the option of surrendering if my bluff is called. If, however, humans are imperfectly able to lie to each other about what sort of people they are—as seems to be the case—then the best available option may be to really be tough, despite the risk that I will occasionally find myself forced to fight when I would be better off surrendering.

None of this argument depends on moral sanctions. I may (indeed do) believe that the tax collector is morally equivalent to the robber. I accept one and fight the other because of my beliefs about other people's behavior—what they will or will not fight for—and because there are beliefs about my behavior which I wish others to hold. We are bound together by a set of mutually reinforcing strategic expectations.

II. TWO ROUTES FROM HOBBS TO HERE

My argument so far has dealt with two ends of an extended process. I started with an explanation of how it was possible, in a two-person world, to take the first steps toward bargaining out of a Hobbesian state

¹³ Rudyard Kipling, "Danegeld," in *Rudyard Kipling's Verse: Definitive Edition* (Garden City: Doubleday, 1940), pp. 716–17.

¹⁴ Robert Frank, *Passions within Reason: The Strategic Role of the Emotions* (New York: Norton, 1988). See also the discussion in Friedman, *Price Theory*, pp. 288–90, which deals with the same strategy seen from the standpoint of the aggressor, the bully committing himself to carry through his threats even if doing so is not in his immediate interest.

¹⁵ Danegeld as it actually existed did not entirely fit the pattern of behavior I am discussing here. The initial Danish invasions were for land and loot. In agreeing to be bought off, the invading armies may have been giving up an opportunity to do something they in fact wanted to do.

of nature. I ended with an explanation of how the same logic maintains civil order as we know it. Missing is any explanation of the intermediate steps by which the complicated and functional order in which we live might have been constructed.

One possibility is legislation. If an important part of the way in which individuals classify actions is "legal/illegal," then the fact of legal change, whether by a king, legislature, or court system, changes the way in which they classify the alternatives, which in turn changes the set of Schelling points. If the court has recognized property rights in water but not in air, I classify pollution of my section of the river as aggression and fight it, by legal, social, or even illegal means. I classify pollution of my air by my neighbor's soap factory as an inconvenient nuisance and either put up with it or try to buy him off. Under these circumstances legislation is, to a considerable degree, self-enforcing; the pattern of property rights might well survive even if the enforcement arm of the state vanished or became impotent.

While this may be part of the explanation for civil order, it cannot be all of it, for at least three reasons. First, some rights have no legal rules associated with them—the right not to have someone cut ahead of me in a line, for example. Second, many, perhaps most, people are selective about which legal rules they take seriously—as can easily be observed on any U.S. highway. And finally, there are well-documented situations in which property rights exist and are respected even though they are inconsistent with the relevant legal rights.

This final point brings up a second possible explanation of how the pattern of expectations might have come into existence—that it is due not to the creation of laws but to the evolution of norms. Robert Ellickson, in a recent book, describes how relations among neighbors function in Shasta County, California.¹⁶ One of his most striking observations is that in many cases, including conflicts over trespass by animals and the allocation of the cost of building fences between neighbors, the inhabitants ignore the relevant laws and act instead according to well-understood nonlegal norms. Ellickson offers no adequate account of how such norms develop or of why they provide, in some contexts but not in all, at least approximately efficient rules. A possible answer to that puzzle brings us back to the two-person social contract discussed in the previous section.¹⁷

One might try to explain functional norms by evolution. Perhaps, over time, societies with better norms conquer, absorb, or are imitated by societies with worse norms, producing a world of well-designed societies. The problem with that explanation is that such a process should take centuries, if not millennia—which does not fit the facts as Ellickson reports

¹⁶ Robert Ellickson, *Order without Law* (Cambridge: Harvard University Press, 1991).

¹⁷ See my review of Ellickson: David Friedman, "Less Law than Meets the Eye," *Michigan Law Review*, vol. 90, no. 6 (May 1992), pp. 1444-52.

them. Whaling norms in the nineteenth century, for example, seem to have adjusted rapidly to changes in the species being hunted.

Perhaps what is happening is evolution, but evolution involving groups much smaller and more fluid than entire societies. Consider a norm, such as honesty, that can profitably be followed by small groups within a society, applicable only within the group. Groups with efficient norms will prosper and grow by recruitment. Others will imitate them. Groups with similar norms will tend to fuse, in order to obtain the same benefits on a larger scale. If one system of norms works better than its competitors, it will eventually spread through the entire society. When circumstances change and new problems arise, the process can repeat itself on a smaller scale, generating modified norms to deal with the new problems. In effect, what we have is the pairwise contracting out of the Hobbesian state of nature, repeated many times between pairs and within small groups.

This conjecture about how norms arise and change suggests a prediction: Even if a norm is efficient, it will not arise if its benefits depend on its being generally adopted. Suppose we define a norm as *locally efficient* if, with regard to any two individuals following the norm, there is no different norm such that at least one would be better off and the other no worse off if they both switched to it. A norm is *globally efficient* if there is no different norm such that at least one person would be better off and nobody worse off if everyone switched to it.¹⁸

Consider the whaling norms that Ellickson discusses. It is in the interest of any pair of captains to agree in advance to an efficient rule for dealing with whales that one ship harpoons and another one brings in, just as it is in the interest of a pair of individuals to agree to be honest with each other. But a rule for holding down the total number of whales killed so as to preserve the population of whales is useful only if almost everyone follows it. The former type of norm existed, the latter did not—with the result that nineteenth-century whalers did an efficient job of hunting one species after another to near-extinction.¹⁹

¹⁸ I apologize to mathematicians and my fellow economists for using “local” and “global” in senses that may seem inconsistent with their usual usage in classifying maxima; I will be happy to consider suggestions for alternative terminology. To make my usage of the terms seem less idiosyncratic, think of a change in norms involving only two people as a small change and one involving many as a large change, and think of an “improvement” as a change to a situation that is Pareto superior for those who are changing. Then a locally efficient set of norms, like a local optimum, is one that cannot be improved by a small change.

For a more mathematically elaborate approach to defining small and large changes and using them to analyze the evolution of rules in a population, see Friedman, “Bilateral Monopoly” (*supra* note 11).

¹⁹ A slightly different way of putting this argument is in terms of what Richard Dawkins has described as the evolution of “memes”: ideas evolving in an environment consisting of the minds of humans. See Richard Dawkins, *The Selfish Gene* (New York: Oxford University Press, 1976), pp. 203–15. One reason a meme—such as the belief that “one ought to be honest toward honest people”—will spread is that those holding it are observed to be more successful as a result. But in order for the process to get past the early stage, when the meme

The evolution of norms, then, provides a second possible account of how we get from Hobbes to here. Where the recognition of rights between two people, such as neighbors, or within a small group, provides mutual benefits, it is in the interest of the parties concerned to recognize such rights.²⁰ By doing so they change the pattern of Schelling points that determines the equilibrium of their interaction, in a way which provides (some) protection for the rights in question. Over a long period of time, the result is to create a set of consistent mutual expectations, and one that tends to be locally, although not necessarily globally, efficient.

III. LAW, JUSTICE, AND EFFICIENCY

In thinking about issues of rights, I find myself playing two quite different roles. As a human being and (like all human beings) an amateur philosopher, I have moral intuitions; from that standpoint, the question I ask is "Why ought one not to steal?" and the answer is "Because it is wicked." As an economist, I ask and answer different questions. One is "What are the consequences of people being free to steal?" Much of the economic analysis of law is devoted to answering questions of that sort. Another is "Why do people (often) not steal?"

This essay is an attempt to answer that final sort of question. I have tried to answer the economist's question about rights rather than the philosopher's, not because economics is more important than moral philosophy, but because I am more confident in my ability to use economics to produce answers.²¹ I have been encouraged in this policy by a curious and convenient coincidence: in most cases, the rules I conclude to be efficient are also the rules I believe to be just.

It is not a double but a triple coincidence. The rules I believe to be efficient and just are also, to a significant degree, the rules enforced by the laws and norms of the society I live in.²² In this essay, I have sketched some ideas about the nature of those rules and how they have evolved.

is still rare in the population, it must be useful to hold the meme even when most other people do not. This works for memes representing norms such as honesty, but it does not work for a meme for conservation of whales. It is accordingly puzzling that memes in favor of conservation have spread so rapidly in the current U.S. population—to the point where belief in conservation has become very nearly the secular equivalent of a state religion.

²⁰ To put the argument in something closer to conventional-game-theory terms, I am combining a familiar concept of dominance in multiplayer games with the idea of transaction costs. Outcome *A* dominates outcome *B* if there is some set of people who together can produce *A* and who all prefer *A* to *B*. A small number of people can produce a change—such as the adoption of a norm among themselves—that only involves changing their own behavior. A large group, because of transaction costs, cannot.

²¹ This point is discussed in David Friedman, *The Machinery of Freedom*, 2d ed. (La Salle, IL: Open Court, 1989), chs. 41–43.

²² The correspondence is not perfect; it would be in some ways closer if I were writing this essay a century ago. A particular form of the correspondence—the claim that the common law tends to be economically efficient—has been a central element of the work of Judge Richard Posner, one of the leading scholars in the law-and-economics tradition. See Richard Posner, *Economic Analysis of Law*, 4th ed. (Boston: Little Brown, 1992).

This raises the question of why, if my account is correct, the rules produced in this way resemble those that I deduce to be efficient and intuit to be just.

In trying to answer that question, I find it useful to start by considering a class of property which underlies all other property and exists even in a Hobbesian state of nature.

I can control the motions of my body by a simple act of will. You can control its motions by imposing overwhelming force, by making believable threats to which I will yield, or in various other ways. Controlling it may be possible for both of us, but it is much cheaper and easier for me. In this sense, we may describe my body as my natural property. The same description applies to my gun—because I know where I hid it and you do not. Even land may be natural property to some extent if my detailed knowledge of the terrain makes it easier for me to use or defend it. Such property is natural inasmuch as my possession of it exists in the state of nature and is independent of social convention. The fact that I can control certain things more cheaply than you can is technology, not law or morals.

Natural property is a useful starting point for explaining the similarities among what is, what should be, and what would be efficient, because it is relevant to all three.

If the account I have offered is correct, our actual civil order is the result of extended bargaining, based ultimately on natural property. It was my control over my body that made the initial steps out of the state of nature possible. So natural property is relevant to what is—to the existing pattern of laws and norms.

In a world of no transaction costs, any initial allocation of property rights is efficient.²³ In a world with positive transaction costs, the basis for choosing among alternative allocations is the cost of enforcing and changing them. A set of rules in which I own my body and you own yours is superior to one in which each owns the other's body, or each has a half interest in each body, in part because it is so much easier to enforce. So we have a Coasian argument for the relevance of natural property to what is efficient.

This argument also provides a second connection between natural property and what is. My earlier arguments suggest that the evolution of rules tends to move in a direction that is at least locally efficient. If so, and if rules that allocate natural property to its natural owner are efficient, we would expect to observe such rules. Put differently, the argument for local efficiency of evolved norms provides a reason for some similarity between the rules we observe and the rules that are efficient.²⁴

²³ Ronald Coase, "The Problem of Social Cost," *Journal of Law and Economics*, vol. 3 (1960), pp. 1-44.

²⁴ I am not claiming here that slavery is never observed or that it is never efficient. One can imagine circumstances in which the self-enforcing element of contracts would be too weak to enforce the contract that would be superior to slavery. An example is the prisoner

What, if anything, does natural property have to do with what ought to be? That depends on what normative account one accepts. For those of us who accept a libertarian account, in which the underlying right is my right to own myself and whatever I have obtained by voluntary agreement with others who own it, the connection is immediate. Self-ownership is both a moral axiom and a technological fact. Voluntary exchange is both a morally legitimate way of altering the pattern of ownership and, if my account of bargaining from the state of nature is correct, a technologically possible way (although not necessarily the only such) of altering a Schelling point and thus an equilibrium.

We now have the beginning of an explanation of the similarity among actual rules, efficient rules, and just rules. The status of this explanation, and of the fact being explained, is not, however, the same for the relation between the first two as it is for the relation of either to the third.

The existence of rules can be observed and the efficiency of rules can be deduced, at least in principle, from observed technologies and economic theory. Thus, the claim that there is some correspondence between what exists and what is efficient is a positive rather than a normative claim.²⁵

What ought to be, on the other hand, is, at least in this essay, simply a description of my moral intuitions. If I conclude that the rules that would be just are similar to both the rules that exist and the rules that would be efficient, that may simply be evidence that my moral judgments are *ex post* rationalizations of the world I live in or the conclusions of my economic analysis.

One further similarity between the ethics and the social order that I have been discussing is worth mentioning. Both are essentially decentralized. The ethical position makes no attempt to evaluate individuals from above—in terms of their worth in the eyes of God. It consists rather of a description of what obligations each individual has to each other individual.²⁶ The social order, to the extent that it is evolved rather than legis-

of war who is confined because there is no other way of enforcing his agreement to pay for his freedom. A counterexample is the institution of parole, as it in fact developed. See Bruno S. Frey and Heinz Buhofer, "Prisoners and Property Rights," *Journal of Law and Economics*, vol. 31 (April 1988), pp. 19–46.

The argument given here provides a possible explanation for the Posnerian thesis that common law tends to be economically efficient. Common law presumably originated as a set of (locally efficient) norms converted over time into legally enforceable rules. If this interpretation is correct, we would expect common law to have become less efficient over time, since the mechanism that generates the efficiency of norms would not apply to legal rules defined and interpreted by a third party.

²⁵ The obvious example of such a claim is Posner's thesis that the common law tends to be economically efficient. Testing that conjecture is an extraordinarily difficult and complicated project, but it is in principle a positive one. See Posner, *Economic Analysis of Law*.

²⁶ This point is discussed at greater length in David Friedman, "Should the Characteristics of Victims and Criminals Count? *Payne v. Tennessee* and Two Views of Efficient Punishment," *Boston College Law Review*, vol. 34, no. 4 (July 1993).

lated, is a set of rules that exist because it was in the interest of pairs of individuals to abide by them, not because it was decided by someone that they would promote the general good of society.²⁷

IV. CONCLUSIONS

The central project of this essay has been to give an account of rights, especially property rights, that is both amoral and alegal—an account that would explain the sort of behavior we associate with rights even in a world lacking law, law enforcement, and feelings of moral obligation.²⁸ I have tried first to explain how, with no legal system to enforce contracts, it might still be possible to contract out of a Hobbesian state of nature, and then to show how the same analysis can be used to understand in what sense a civil order, such as our own society, is different from a Hobbesian state of nature. Having offered answers to those questions, I then tried to show how we might get from the state of nature to something like the present society, and tried to use the analysis to partially explain the puzzling similarity between actual rules, just rules, and efficient rules.

If my analysis is correct, civil order is an elaborate Schelling point, maintained by the same forces that maintain simpler Schelling points in a state of nature. Property ownership²⁹ is alterable by contract because Schelling points are altered by the making of contracts. Legal rules are in large part a superstructure erected upon an underlying structure of self-enforcing rights.

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²⁷ Which is not to imply that legislated rules necessarily are designed for the good of society, but only that they might be.

²⁸ This is not the only way in which a positive account of rights might be developed. Sugden pursues a similar project in a framework where conventions develop as players of a multiplayer game learn by experience and alter their behavior accordingly; see Sugden, *Economics of Rights, Co-operation, and Welfare* (*supra* note 3). Another alternative would be a sociobiological explanation, in which respect for property is an inherited behavior pattern that evolved because it resulted in increased fitness—which is to say, reproductive success.

It is unclear to what extent these are really competing explanations, rather than different views of the same elephant. The sociobiological account may be seen as Sugden's repeated game, repeated through many generations, or as my account in a world where humans' ability to alter property rights by contract simply reflects the richer set of Schelling points available to them. My sketchy account of the development of norms and Sugden's more detailed account of the development of conventions are the same, save that he emphasizes anonymous adaptation where I emphasize explicit contracts among small subgroups. His account of the underlying games takes the available strategy set as given, where mine takes it as a subjective fact about the players that may, under some circumstances, be alterable. The similarities among the explanations are close; it is unclear whether the differences are sufficiently great to lead to significant differences in their implications.

²⁹ By "property ownership" I mean the ability to control things, not the legal right to do so.